TABLE IX

Transfer-inclusive timings and resident/active execution evidence for LARGE FP16 cases (all values in milliseconds). M: measured; P: profile-derived estimate; C: compiler-reported schedule estimate; D: derived native-cycle budget. Distinct timing scopes are retained rather than pooled into a common speedup.

		NVIDIA A100			Groq		Graphcore IPU		Cerebras CS-3			
Kernel	Case	I/O+ M	Resident M	Active P	I/O+ M	Schedule C	I/O+ M	Active P	Stream C	First C	Native D	Native+ctx D
BICG	c1	0.4921	0.099328	0.03638	1.1245	0.12061	2.613	0.11662	–	–	–	–
	c2	0.545	0.13875	0.10286	1.0585	0.055999	2.4901	0.016811	3.1334	4.6355	0.011681	0.18368
	c3	1.9801	1.5508	1.3474	1.1744	0.17106	2.5353	0.016215	3.1334	4.6355	0.011681	0.18368
MVT	c1	0.53198	0.10906	0.04384	1.1029	0.048283	2.5716	0.068494	3.2704	4.7629	0.014765[b]	0.30143[b]
	c2	0.54022	0.10291	0.03726	1.0897	0.048283	2.5536	0.068494	6.9289	6.9863	0.082731	0.42673
	c3	0.60674	0.16128	0.090139	1.1336	0.061336	2.538	0.026278	3.2704	4.7629	0.014765[b]	0.30143[b]
GESUMMV	c1	0.4368	0.098816	0.03554	0.89706	0.034023	2.0517	0.0070254	3.8044	3.8045	0.004116[b]	0.17612[b]
	c2	0.46472	0.10138	0.04742	0.89382	0.030992	2.0637	0.0077714	3.8044	3.8045	0.004116[b]	0.17612[b]
GEMM	c2	0.65502	0.14848	0.08684	1.9243	0.078151	5.5945	0.068494	–	–	–	–
	c3	0.67872	0.15667	0.1027	2.0752	0.078151	5.6157	0.068495	6.8837	6.8837	0.045416[b]	0.27475[b]
IIR filter bank	c1	202.17	202.1	237.41[a]	0.05143	0.014652	–	–	–	–	–	–
Lorenz RK4	c3	15.576	17.186	–	–	–	–	–	102.76	102.82	10.138	102.67
Low-rank	c1	2.9731	2.7448	2.3519	–	–	4.9961	0.62482	–	–	–	–
	c2	2.1075	1.877	1.5849	–	–	8.7854	0.41413	–	–	–	–
Poisson CG	c1	0.7391	0.6999	0.83097	0.08626	0.066423	0.35793	0.10574	–	–	–	–
DOITGEN	c1	0.73928	0.091648	0.03964	2.444	0.10227	20.983	0.019367	11.795	11.795	0.070929	0.1856
	c3	4.1502	3.5471	3.5872	2.3144	0.10227	20.918	0.019367	19.007	19.007	0.015199	0.12987
SYR2K	c1	0.74724	0.19558	0.15446	3.0047	0.10915	6.9695	0.11696	–	–	–	–
	c2	0.74486	0.19866	0.17588	3.0178	0.11848	6.7797	0.091851	–	–	–	–
TRMM	c1	0.51442	0.13978	0.087739	1.976	0.059546	10.378	0.14558	–	–	–	–
	c3	5.8306	8.8325	5.0293	1.9488	0.057678	–	–	–	–	–	–

Timing boundaries. I/O+ is a measured serialized host-to-host call, including input/output transfers and invocation, with setup excluded. A100/Groq use PCIe; Graphcore uses the deployed IPU fabric. A100 Resident is CUDA-event elapsed time for a captured graph with device-resident tensors (including HBM access); Active is the union of profiled kernel/device-memory intervals with idle gaps excluded. Groq Schedule is the successful executable's compiler cycle estimate. IPU Active reconstructs compute, internal exchange and internal synchronization phases.

CS-3 estimates. Stream is the predicted steady-state interval with modeled streaming stalls; First is the predicted first compiled iteration. Neither is a measured host-to-host call. Native sums the compiler's wafer-native operation cycles assuming preloaded data/layouts, resident code and serial operations; Native+ctx also retains the original context-switch cycle budget. Both D columns are conditional component budgets, not predictions from a newly compiled resident executable or proof of memory-placement feasibility.

[a] IIR profiling perturbs timing; its active estimate exceeds the same process's capture-disabled resident interval and is diagnostic only. [b] CS-3 has uncosted host helpers (MVT addition; GESUMMV/GEMM reduction/distribution), so the D columns omit required remapping costs. Independent runs and instrumentation can also make active estimates exceed archived elapsed medians; differences are not isolated transfer costs. – denotes unavailable qualified timing evidence, not zero latency. Case IDs and source hashes are recorded in the accompanying CSV.

their paired unpatched sources: four Luna, seven Terra, and nine SOL implementations across five kernels. Each receives twelve deterministic holdouts: four local perturbations at two magnitudes, two independent random inputs, half- and double-scale inputs, two structured fields, and two sparse interventions. Workload dimensions, iteration counts, and fixed parameters are unchanged. Sources, C contracts, and input policies are frozen; no repair uses holdout feedback.

Both versions receive identical inputs on the A100. Expected outputs come from the original C computation with inputs injected after initialization and dependent state rebuilt consistently. We report the repair loop's worst-component relative L_2 error in percent:

$$e = 100 \max_j \frac{\|\widehat{y}_j - y_{C,j}\|_2}{\max(\|y_{C,j}\|_2, 10^{-30})}, \tag{1}$$

where j indexes independently scored C outputs, including scalars. Finite minima, medians, and maxima exclude unavailable numerical errors; their counts accompany every summary in Table X. NaN/Inf outputs receive no arbitrary percentage. Repeated forwards and input/dtype audits accompany the measurements; all 260 unpatched FP64 checks, including canonical controls, satisfy the C semantic check.

Across 240 implementation–input holdouts, unpatched FP16 sources produce 196 NaN outputs and 44 finite outputs. Repair makes 187 of those previously non-finite observations finite, yielding 231 finite outputs, seven Inf outputs, and two NaN outputs. Source-bound replays confirm the non-finite values, including NaNs whose repeating locations triggered the original strict repeatability rejection. These changes demonstrate transferred improvements in numerical stability; finiteness alone does not establish accuracy.

Jacobi-2D provides a fully finite accuracy comparison. Unpatched errors span 4.266–12.605%, with median 12.482%; repaired errors span 0.0238–1.042%, with median